

Teaching Exam

Subject – Computer Science

Topic – DBMS

Lecture No.- 11



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Topics to be Covered

Topic



- 1:- Practice on Closer Method ✓
- 2:- Functional Dependency ✓
- 3:- Types of Functional Dependency ✓
- 4:- Trivial Function Dependency ✓
- 5:- Non-Trivial Functional Dependency ✓
- 6-Multivalued Functional Dependency ✓
- 7- Transitive Functional Dependency ✓
- 8- Full and Partial Functional Dependency ✓
- 9- Conclusion. ✓
- 10->2-Normal Form ✓



Topic : Practice on Closer Method

The maximum number of superkeys for the relation schema $R(E, F, G, H)$ with E as the key is

(A) 5

(B) 6

(C) 7

(D) 8

$R(E, F, G, H)$

key

E, F as the key

$n = \text{Total no of Relation}$

$$2^{(n-1)}$$

$$2^{(4-1)}$$

$$2^3 = 8$$

(Total no Attribute - no of Attribute under key formation)

$$2^{4-1}$$



Topic : Practice on Closer Method



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Consider a relation $R(A, B, C, D, E)$ with the following three functional dependencies.

$AB \rightarrow C$; $BC \rightarrow D$; $C \rightarrow E$;

The number of superkeys in the relation R is

Can't $\rightarrow \{A\}$
Super Key $\rightarrow \{AB\}$

$$2^{(5-2)} \\ 2^3 \Rightarrow 8$$

$$\begin{aligned} \overline{AB}^+ &= \{A, B, C, D, E\} & A^+ &= \{X\} & A^+ &= \{A\} \\ & & B^+ &= \{X\} & C^+ &= \{C, E\} \\ & & D^+ &= \{X\} & E^+ &= \{E\} \end{aligned}$$



Topic : Practice on Closer Method



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Second Method to be solved

Only one candidate key is possible for given relation = AB

Superkeys of the relation = $2^{\text{Total number attributes in relations} - \text{Total number of attributes in each key}}$

Superkeys of the relation = 2^{5-2}

Superkeys of the relation = $2^3 = 8$ keys



Topic : Practice on Closer Method

Relation R has eight attributes ABCDEFGH. Fields of R contain only atomic values. $F = \{CH \rightarrow G, A \rightarrow BC, B \rightarrow CFH, E \rightarrow A, F \rightarrow EG\}$ is a set of functional dependencies (FDs) so that F^+ is exactly the set of FDs that hold for R.

$$F^+ = \{F, D\}$$

Q- How many Candidate key is possible ?

(a) 3

(b) 4

(c) 5

(d) 6

$\{AD\}^+ \times$
 $\{BD\}^+ \times$

$C^+ = \{C\}$

non-prim

Prime Attribute

$\{E, D\}$

$\rightarrow$ Superkey

$\{E, D\}$

$\rightarrow$ Superkey

$\{H \rightarrow G$
 $\{A \rightarrow BC\}$

$B \rightarrow CFH$

$E \rightarrow A$

$F \rightarrow EG$

$\{ED\}^+ = \{C, E, D, A, B, C, F, H, G\}$

$\{AD\}$
 $\{BD\}$
 $\{ED\}$
 $\{FD\}$
 $\{CFD\}$
 $\{CED\}$

4

$\rightarrow$ Superkey

$\rightarrow$ Superkey



Topic : Practice on Closer Method

$$\checkmark AB \rightarrow AB$$

5. In a schema with attributes A, B, C, D and E, following set of functional dependencies are given:

A $\rightarrow$ B

$$B D^+ = \{B, D\}$$

A $\rightarrow$ C

$$C D^+ = \{C, D, E, (A)B\}$$

CD $\rightarrow$ E

B $\rightarrow$ D

$$B C^+ = \{B, (C), D, E, A\}$$

$$A C^+ = \{A, (C), (B), D, E\}$$

E $\rightarrow$ A

$$A \rightarrow B$$

B is determined by A
Non-Trivial functional
depend

Which of the following functional dependencies is NOT implied by the above set?

(a) CD $\rightarrow$ AC

(b) BD $\rightarrow$ CD

(c) BC $\rightarrow$ CD

(d) AC $\rightarrow$ BC



Topic : Functional Dependency



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A functional dependency occurs when one attribute uniquely determines another attribute within a relation. It is a constraint that describes how attributes in a table relate to each other. If attribute A functionally determines attribute B we write this as the $A \rightarrow B$.

(Determinate) $A \twoheadrightarrow B$ (Dependent Attribute)

A Determinate B

OR

B is Determined by A

EX-

$SID \twoheadrightarrow SNAME$

1	RAHUL
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Topic : Functional Dependency

Functional dependencies are used to mathematically express relations among database entities and are very important to understanding advanced concepts in Relational Database Systems.

Example:

roll_no	name	dept_name	dept_building
42	abc	CO	A4
43	pqr	IT	A3
44	xyz	CO	A4
45	xyz	IT	A3
46	mno	EC	B2
47	jkl	ME	B2



Topic : Functional Dependency

From the above table we can conclude some valid functional dependencies:

- $\text{roll_no} \rightarrow \{\text{name}, \text{dept_name}, \text{dept_building}\}$ Here, roll_no can determine values of fields name, dept_name and dept_building, hence a valid Functional dependency
- $\text{roll_no} \rightarrow \text{dept_name}$, Since, roll_no can determine whole set of {name, dept_name, dept_building}, it can determine its subset dept_name also.
- $\text{dept_name} \rightarrow \text{dept_building}$, Dept_name can identify the dept_building accurately, since departments with different dept_name will also have a different dept_building
- More valid functional dependencies: $\text{roll_no} \rightarrow \text{name}$, $\{\text{roll_no}, \text{name}\} \rightarrow \{\text{dept_name}, \text{dept_building}\}$, etc.

Here are some invalid functional dependencies:

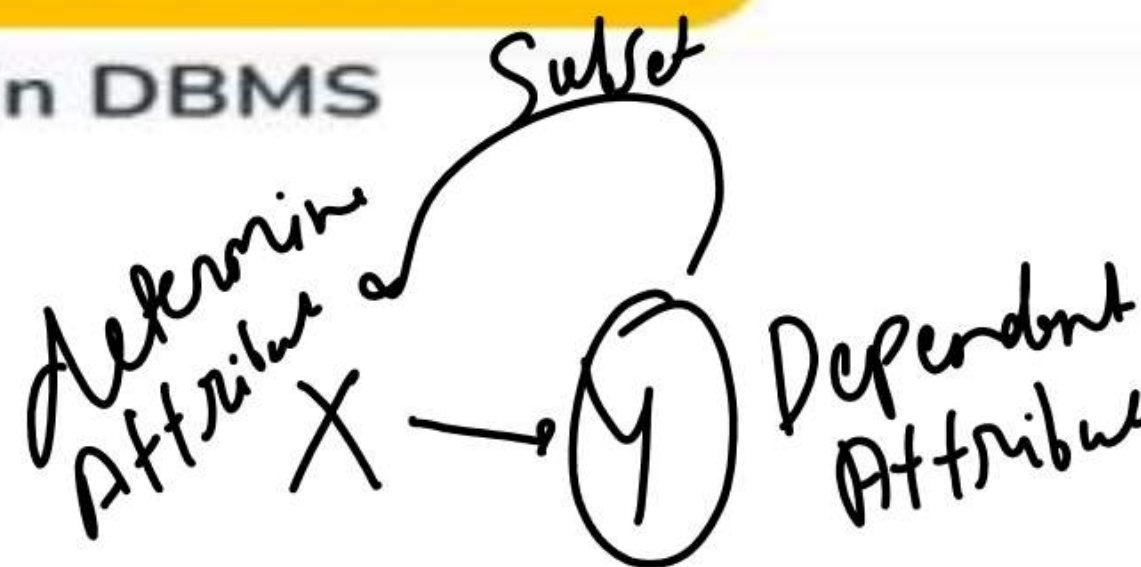
- $\text{name} \rightarrow \text{dept_name}$ Students with the same name can have different dept_name, hence this is not a valid functional dependency.
- $\text{dept_building} \rightarrow \text{dept_name}$ There can be multiple departments in the same building. Example, in the above table departments ME and EC are in the same building B2, hence $\text{dept_building} \rightarrow \text{dept_name}$ is an invalid functional dependency.
- More invalid functional dependencies: $\text{name} \rightarrow \text{roll_no}$, $\{\text{name}, \text{dept_name}\} \rightarrow \text{roll_no}$, $\text{dept_building} \rightarrow \text{roll_no}$, etc.



Topic :Types of Functional Dependency

Types of Functional Dependencies in DBMS

1. Trivial functional dependency ✓
2. Non-Trivial functional dependency ✓
3. Multivalued functional dependency ✓
4. Transitive functional dependency ✓



1. Trivial Functional Dependency

In Trivial Functional Dependency, a dependent is always a subset of the determinant. i.e. If $X \rightarrow Y$ and Y is the subset of X , then it is called trivial functional dependency.

Symbolically: $A \rightarrow B$ is trivial functional dependency if B is a subset of A .

The following dependencies are also trivial: $A \rightarrow A$ & $B \rightarrow B$

Example 1 :

- $ABC \rightarrow AB$,
- $ABC \rightarrow A$
- $ABC \rightarrow ABC$



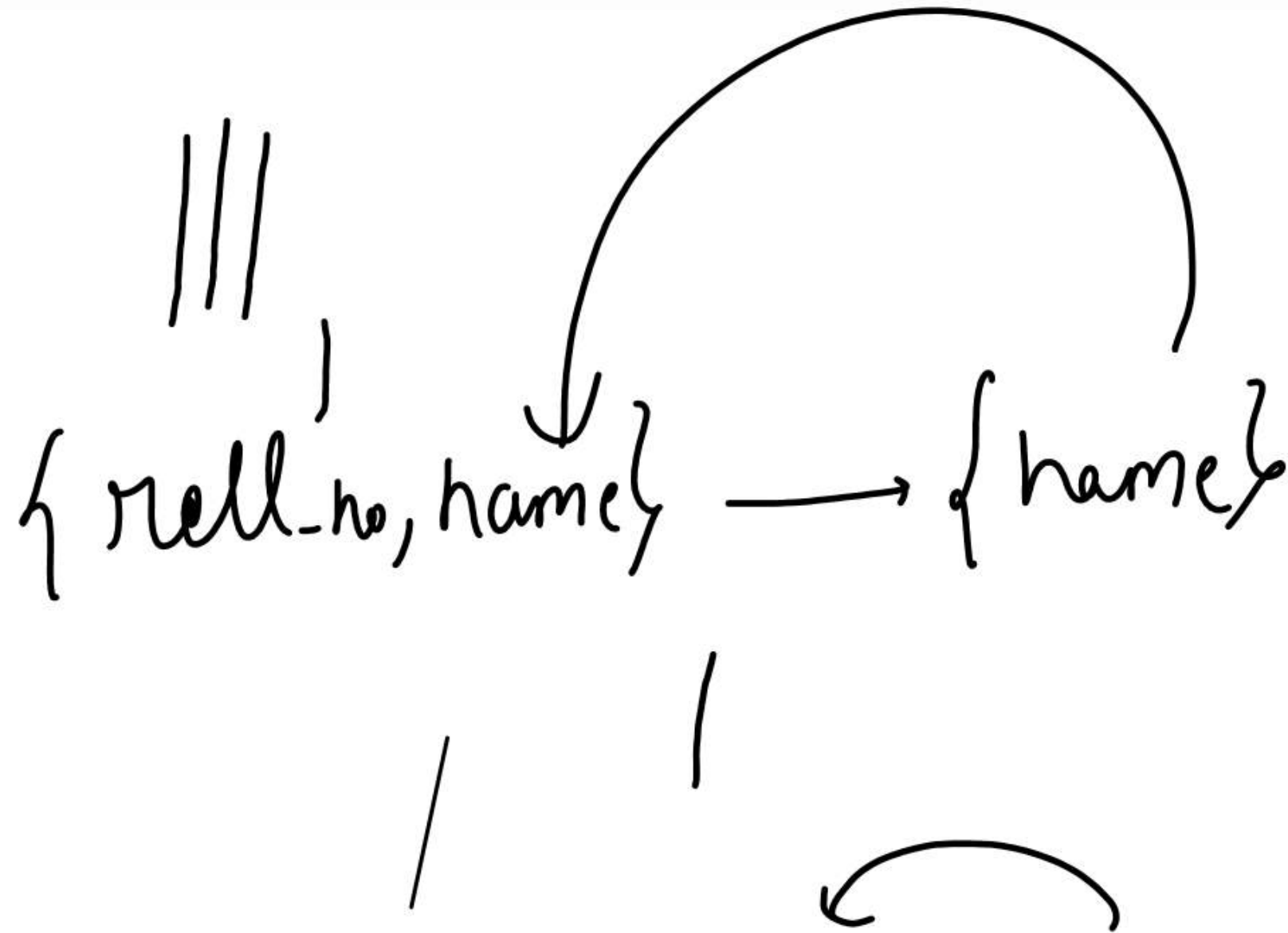
Topic : Trivial Functional Dependency

Example 1 :

- $ABC \rightarrow AB$
- $ABC \rightarrow A$
- $ABC \rightarrow ABC$

Example 2:

roll_no	name	age
42	abc	17
43	pqr	18
44	xyz	18



Here, $\{\text{roll_no}, \text{name}\} \rightarrow \text{name}$ is a trivial functional dependency, since the dependent name is a subset of determinant set $\{\text{roll_no}, \text{name}\}$. Similarly, $\text{roll_no} \rightarrow \text{roll_no}$ is also an example of trivial functional dependency.



Topic : Non-Trivial Functional Dependency



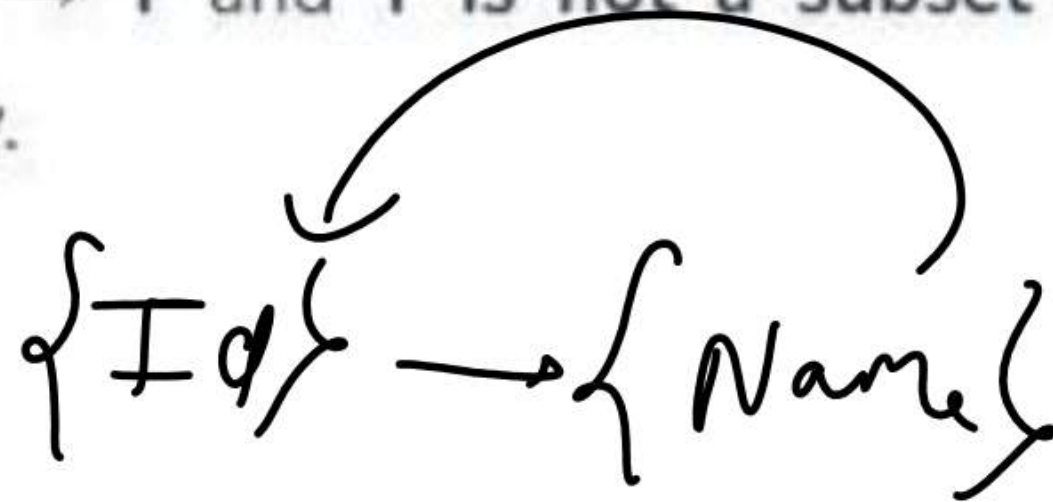
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2. Non-trivial Functional Dependency

In Non-trivial functional dependency, the dependent is strictly not a subset of the determinant. i.e. If $X \rightarrow Y$ and Y is not a subset of X , then it is called Non-trivial functional dependency.

Example 1 :

- $Id \rightarrow Name$
- $Name \rightarrow DOB$

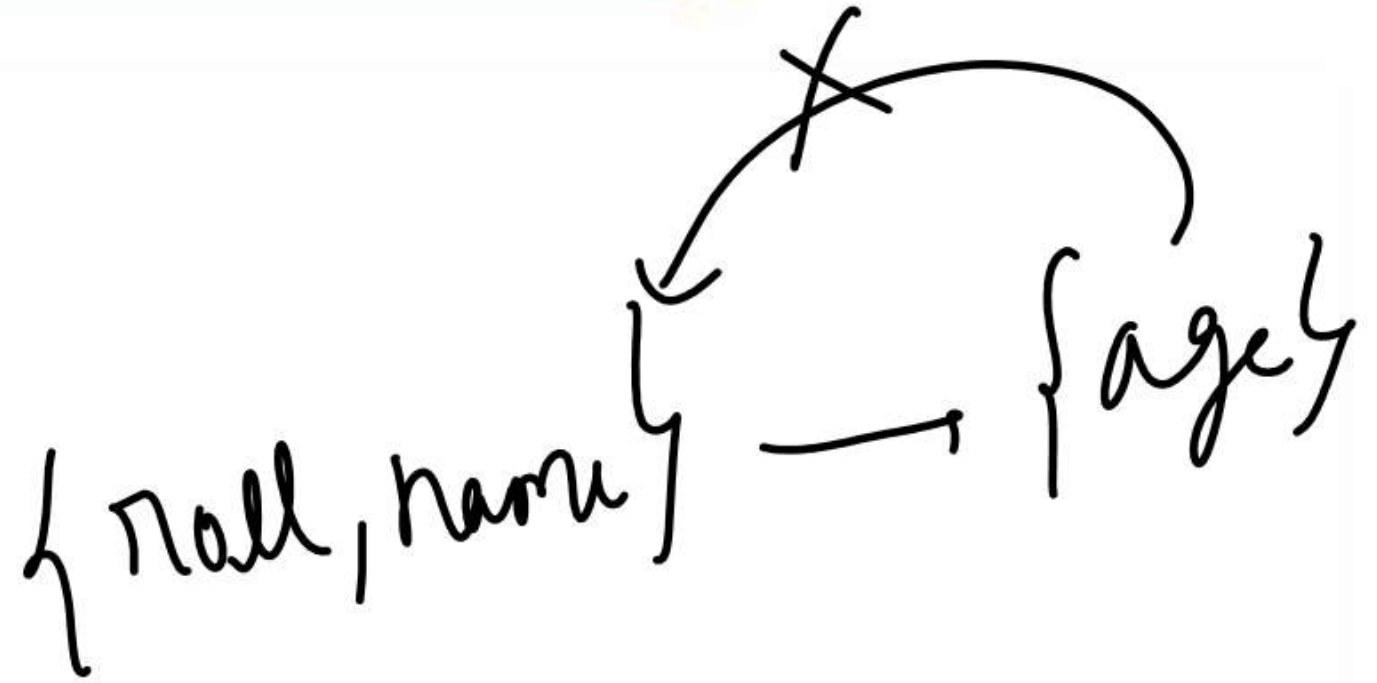




Topic : Non-Trivial Functional Dependency

Example 2:

roll_no	name	age
42	abc	17
43	pqr	18
44	xyz	18



Here, $roll_no \rightarrow name$ is a non-trivial functional dependency, since the dependent name is not a subset of determinant $roll_no$. Similarly, $\{roll_no, name\} \rightarrow age$ is also a non-trivial functional dependency, since age is not a subset of $\{roll_no, name\}$.



Topic : Multivalued Functional Dependency



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4. Multivalued Functional Dependency

In Multivalued functional dependency, entities of the dependent set are not dependent on each other. i.e. If $a \twoheadrightarrow \{b, c\}$ and there exists no functional dependency between b and c , then it is called a multivalued functional dependency.

Example:

bike_model	manuf_year	color
tu1001	2007	Black
tu1001	2007	Red
tu2012	2008	Black
tu2012	2008	Red



Topic : Multivalued Functional Dependency



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4. Multivalued Functional Dependency

In Multivalued functional dependency, entities of the dependent set are not dependent on each other. i.e. If $a \twoheadrightarrow \{b, c\}$ and there exists no functional dependency between b and c , then it is called a multivalued functional dependency.

Example:

bike_model	manuf_year	color
tu1001	2007	Black
tu1001	2007	Red
tu2012	2008	Black
tu2012	2008	Red



Topic :Multivalued Functional Dependency



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In this table:

- X: bike_model
- Y: color
- Z: manuf_year

For each bike model (bike_model):

1. There is a group of colors (color) and a group of manufacturing years (manuf_year).
2. The colors do not depend on the manufacturing year, and the manufacturing year does not depend on the colors. They are independent.
3. The sets of color and manuf_year are linked only to bike_model.

That's what makes it a multivalued dependency.

In this case these two columns are said to be multivalued dependent on bike_model.

These dependencies can be represented like this:



Topic : Transitive Functional Dependency

In transitive functional dependency, dependent is indirectly dependent on determinant. i.e. If $a \rightarrow b$ & $b \rightarrow c$, then according to axiom of transitivity, $a \rightarrow c$. This is a transitive functional dependency.

Example:

enrol_no	name	dept	building_no
42	abc	CO	4
43	pqr	EC	2
44	xyz	IT	1
45	abc	EC	2

Here, $\text{enrol_no} \rightarrow \text{dept}$ and $\text{dept} \rightarrow \text{building_no}$. Hence, according to the axiom of transitivity, $\text{enrol_no} \rightarrow \text{building_no}$ is a valid functional dependency. This is an indirect functional dependency, hence called Transitive functional dependency.



Topic : Full and Partial Functional Dependency



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6. Fully Functional Dependency

In full functional dependency an attribute or a set of attributes uniquely determines another attribute or set of attributes. If a relation R has attributes X, Y, Z with the dependencies $X \rightarrow Y$ and $X \rightarrow Z$ which states that those dependencies are fully functional.

7. Partial Functional Dependency

In partial functional dependency a non key attribute depends on a part of the composite key, rather than the whole key. If a relation R has attributes X, Y, Z where X and Y are the composite key and Z is non key attribute. Then $X \rightarrow Z$ is a partial functional dependency in RBDMS.



Topic :Conclusion on Functional Dependency



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Conclusion

Functional dependency is very important concept in database management system for ensuring the data consistency and accuracy.



Topic : 2-NF

What is Second Normal Form (2NF)?

Second Normal Form (2NF) is based on the concept of fully **functional dependency**. It is a way to organize a database table so that it reduces **redundancy** and ensures **data consistency**. For a table to be in 2NF, it must first meet the requirements of **First Normal Form (1NF)**, meaning all columns should contain **single**, indivisible values without any repeating groups. Additionally, the table should not have **partial dependencies**.

The primary goal of **Second Normal Form** is to eliminate **partial dependencies**. A **partial dependency** happens when a non-prime attribute (an attribute not part of a candidate key) depends on only a part of a composite primary key, rather than on the entire key. Removing these partial dependencies helps in reducing redundancy and preventing update anomalies.



Topic : 2-NF



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Example of Second Normal Form (2NF)

Consider a table storing information about students, courses, and their fees:

COURSES Table

STUD_NO	COURSE_NO	COURSE_FEE
1	C1	1000
2	C2	1500
1	C4	2000
4	C3	1000
4	C1	1000
2	C5	2000



Topic : 2-NF



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- There are many courses having the same course fee. Here, COURSE_FEE cannot alone decide the value of COURSE_NO or STUD_NO.
- COURSE_FEE together with STUD_NO cannot decide the value of COURSE_NO.
- COURSE_FEE together with COURSE_NO cannot decide the value of STUD_NO.
- The candidate key for this table is {STUD_NO, COURSE_NO} because the combination of these two columns uniquely identifies each row in the table.
- COURSE_FEE is a **non-prime attribute** because it is not part of the candidate key {STUD_NO, COURSE_NO}.
- But, COURSE_NO $\rightarrow$ COURSE_FEE, i.e., COURSE_FEE is dependent on COURSE_NO, which is a proper subset of the candidate key.
- Therefore, Non-prime attribute COURSE_FEE is dependent on a proper subset of the candidate key, which is a partial dependency and so this relation is not in 2NF.



Topic : 2-NF

STUDENT_COURSES Table

STUD_NO	COURSE_NO
1	C1
2	C2
1	C4
4	C3
4	C1
2	C5

COURSES Table

COURSE_NO	COURSE_FEE
C1	1000
C2	1500
C4	2000
C3	1000
C5	2000

Now, each table is in 2NF:

- The Course Table ensures that **COURSE_FEE** depends only on **COURSE_NO**.
- The Student-Course Table ensures there are no partial dependencies because it only relates students to courses.

Now, the **COURSE_FEE** is no longer repeated in every row, and each table is free from partial dependencies. This makes the database **more efficient** and **easier to maintain**.

THANK YOU